

MID TERM EXAMINATION – Spring 2024
Information Security (InfoSec)

Course Code: SENS 4533

Total marks: 65

Duration: 90 mins

Name: _____

Reg Number: _____

Section: _____

NOTE:

Attempt all questions in the same order on the answer sheet.
Question paper is to be returned with the answer sheets.

Q No. 1: [13 Marks]

Mark as True/False. (Attempt on Answer Sheet)

1. Traffic analysis is an example of an active attack. **T**
2. Columnar transpositions do not affect letter frequency distributions. **F**
3. 3DES with 2 keys, where $K_1 = K_2$, is backward compatible with DES. **T**
4. Consider someone sending a stock sell order to his agent; for this message, confidentiality is more important than accountability. **T**
5. Monoalphabetic ciphers do not change/hide plaintext letter frequencies in ciphertext. **T**
6. Hill cipher can have flatter letter frequency distribution than Playfair cipher. **F**
7. In Hill Cipher, the size of key matrix determines the block size for encryption. **T**
8. Columnar transpositions do not affect letter frequency distributions. **F**
9. During decryption in DES, we use the Inverse Initial Permutation (IP^{-1}) before the IP. **F**
10. Monoalphabetic ciphers are stronger than Polyalphabetic ciphers because frequency analysis is tougher on the former. **F**
11. Counter mode (CTR) can be employed for encrypting data in high-speed links. **T**
12. The cipher that is resistant to chosen text attacks is stronger than the cipher that is safe only against ciphertext only attack. **F**
13. In CBC mode, the current ciphertext is XORed with the previous plaintext block. **T**

Q No. 2: [12 Marks]

Mark the best answer: (Attempt on Answer Sheet)

1. In DES algorithm the round key is _____ bits, and the round function input is _____ bits.

a) 48, 32	b) 64, 32
c) 56, 24	d) 32, 32
2. Which mode of operation in DES encrypts each block of plaintext separately, making it independent of other blocks:

a) ECB (Electronic Codebook)	b) CBC (Cipher Block Chaining)
c) CFB (Cipher Feedback)	d) OFB (Output Feedback)

3. In which mode does DES XOR the previous ciphertext block with the current plaintext block before encryption:
- a) ECB (Electronic Codebook) ☒ b) CBC (Cipher Block Chaining)
 c) CFB (Cipher Feedback) d) OFB (Output Feedback)
4. Identify the technique the Playfair Cipher is based on:
- a) matrix operations b) frequency analysis
☒ c) substitution and permutation d) mathematical functions
5. The S-Box in DES Algorithm performs:
- a) permutation b) substitution
 c) expansion ☒ d) compression
6. If Caesar Cipher with a shift of 10 is used to encrypt the plaintext "SECURITY", select the resulting ciphertext:
- a) COGZYNRP b) KOWDROHC
 c) MOHEBZTM d) KIXYPXNC
7. Identify the number of keys that Triple DES algorithm uses:
- a) 2 ☒ b) 3
 c) 4 d) 2 or 3
8. In RC4, what is the purpose of pseudorandom generation algorithm (PRGA):
- a) key generation ☒ b) generating keystream c) S-box setup d) initialization vector
9. Why is Vernam Cipher considered theoretically unbreakable:
- a) due to its complexity b) because of its simplicity
 c) it uses a large key size d) it requires a special decoding device
10. In Hill Cipher, what type of mathematical operation is performed on plaintext:
- a) substitution b) transposition c) addition ☒ d) matrix multiplication
11. In Vigenère Cipher, what is the key used for:
- a) determining the number of rounds b) shifting the entire alphabet
☒ c) determining the substitution alphabet d) selecting the encryption algorithm
12. In Vigenère Cipher, what is the Kasiski examination used for:
- a) breaking the key b) decoding the message
 c) detecting frequency analysis d) analyzing the key length

Q No. 3: [6+7+6+6 = 25 Marks]

- a. Given ciphertext is obtained after encryption by using Rail Fence transposition cipher. Can you recover the original plaintext. If your answer is YES, find the plaintext; and if your answer is NO, then give the reason. (Hint: Cryptanalysis)

Ciphertext: "CTGODRPOOYNWEGYLKEL"

Handwritten: c 1 2 3 4 5

Hill cipher is vulnerable to known plaintext attack; and the corresponding plaintext and ciphertext pairs are given. Find the 2x2 key matrix which is used for encryption.

Plaintext: HILL

Ciphertext: HMKD

- c. Explain how many brute force attempts are required to break an ATM pin code, if ATM pin is a 4-digit Octal Number.
- d. Given below is Hex dump of a captured Ethernet frame.

0000	52 54 00 de c9 57 00 01 38 21 83 e2 08 00 45 00	RT...W.. B!....E.
0010	00 72 28 b9 40 00 2e 06 58 cb 45 38 c4 ea c0 a8	.r(.0... X.E8....
0020	00 37 00 15 06 08 af cc c0 42 48 90 b4 91 50 18	.7..... .BH...P.
0030	16 d0 bb 84 00 00 20 73 79 73 74 65 6d 20 77 69	 s ystem w/
0040	74 68 6f 75 74 20 61 75 74 68 6f 72 69 74 79 20	thout au thority
0050	6f 73 70 60 60 70 68 70 62 68 72 72 70 6f 66 70	

File: tcpCapture.25 KB (X) DL F: 296 D: 7 M: 0

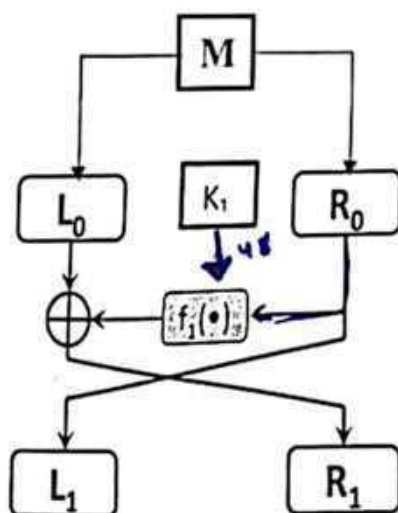
Answer the following questions:

- What is the source & destination IP address.
- How do you check which type of packet is encapsulated in Ethernet frame.
- What is the length of the IP header.

Q No. 4: [15 Marks]

Calculate the value of L_1 and R_1 using the information given below:

Suppose you are given a message (M) in binary = 0000 0001 0010 0011 and the key (K_1) in binary = 0001 1011. The round function f_1 is defined as $f_1 = K_1 \oplus R_0$. Permute the message M using the IP table given below before dividing the message into left and right bits. (Start counting from MSB or left if needed).



IP

16	10	7	1
3	5	6	2
8	11	14	1
4	9	12	